

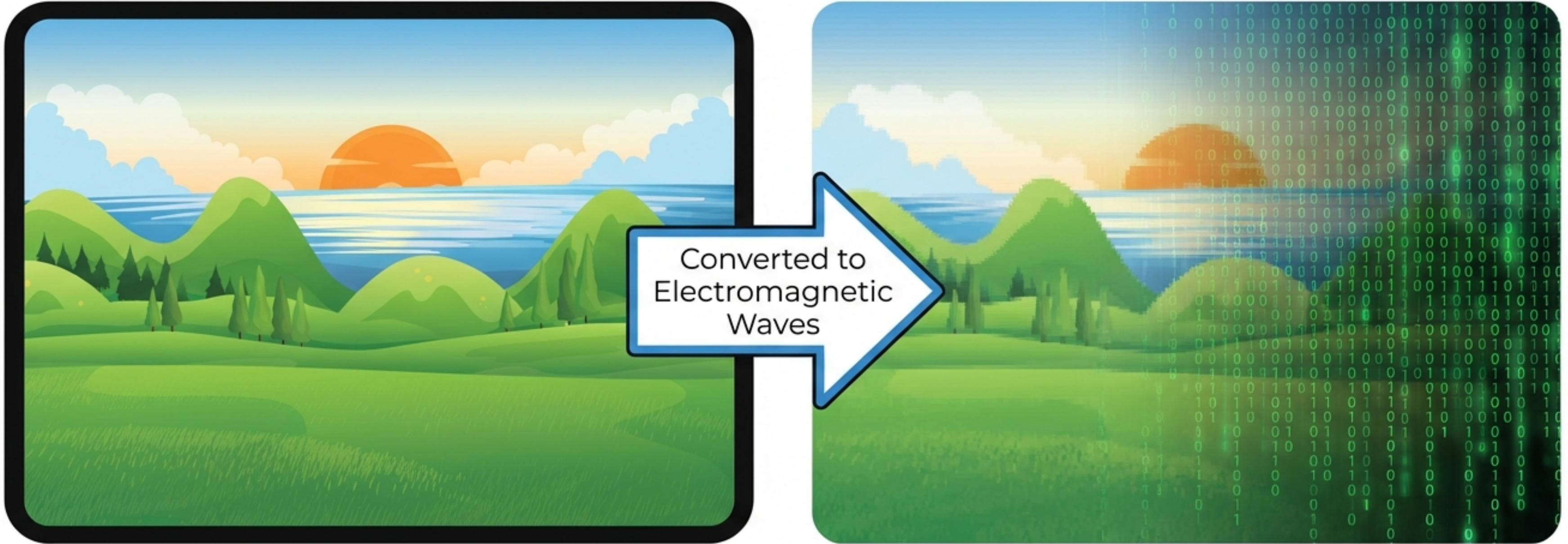
10. Data Transfer

How the Internet Connects Our World

Whether for work or entertainment, we use the internet every day. But have you ever wondered how a click travels around the world?



The Magic Behind the Screen



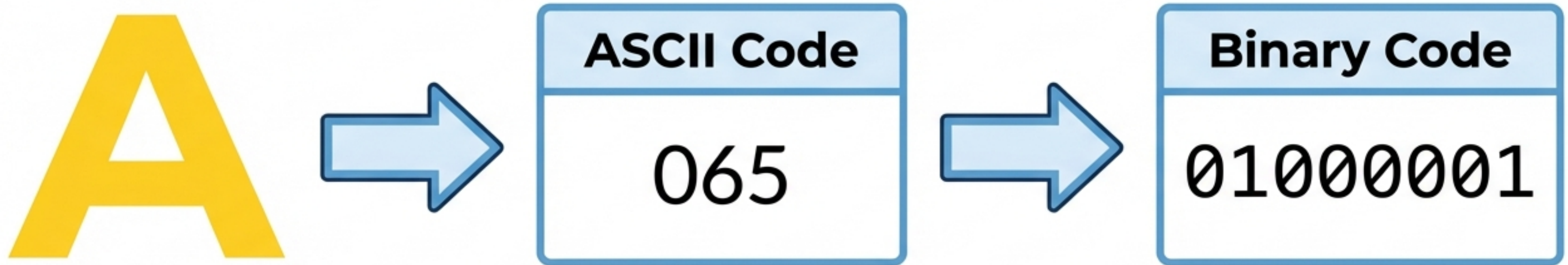
What Humans See

Colors, shapes, and images made of pixels.

What Computers See

Data. Computers only understand code.

Breaking the Code: ASCII & Binary



ASCII (American Standard Code for Information Interchange)

The standard language for encoding text. Every character on your screen is assigned a unique 8-bit code.

Letter	ASCII	Binary	Letter	ASCII	Binary
a	097	01100001	A	065	01000001
b	098	01100010	B	066	01000010
c	099	01100011	C	067	01000011
d	101	01100100	D	068	01000100
e	102	01100110	E	069	01000110
f	102	01100110	F	070	01000110
g	103	01100111	G	071	01000111
h	104	01101000	H	073	01001000
i	105	01101001	I	073	01001001
j	106	01101010	J	074	01001010
k	108	01101011	K	075	01001011
l	109	01101100	L	076	01001100
m	110	01101101	M	077	01001101
n	111	01101110	N	078	01001110
o	111	01101110	O	079	01001111
p	112	01101111	P	080	01001111
q	113	01101111	Q	081	01001111
r	114	01101111	R	082	01001111
s	115	01101111	S	083	01001111
t	116	01101111	T	084	01001111
u	117	01101111	U	085	01001111
v	117	01101111	V	085	01001111
w	118	01101111	W	086	01001111
x	119	01101111	X	087	01001111
y	120	01101111	Y	088	01001111
z	122	01101111	Z	089	01001111

Speaking the Language of Math

Decimal System (Base 10)



Digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
Used by: Humans

Binary System (Base 2)



Digits: 0, 1
Used by: Computers

Base: The number of unique digits used to represent numbers.

How-To: Convert Decimal to Binary

$$156 \div 2 = 78$$

$$78 \div 2 = 39$$

$$39 \div 2 = 19$$

$$19 \div 2 = 9$$

$$9 \div 2 = 4$$

$$4 \div 2 = 2$$

$$2 \div 2 = 1$$

$$1 \div 2 = 0$$

(Remainder 0)

(Remainder 0)

(Remainder 1)

(Remainder 1)

(Remainder 1)

(Remainder 0)

(Remainder 0)

(Remainder 1)

**Read in
Reverse**



10011100

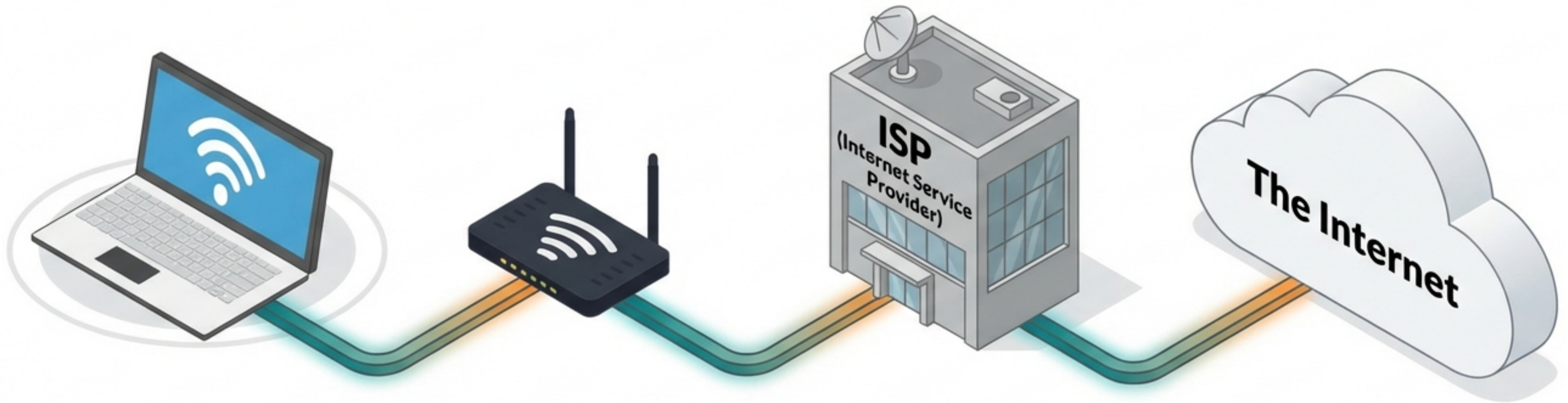
How-To: Convert Binary to Decimal

Powers of 2	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Value	128	64	32	16	8	4	2	1
The Binary Code for 179	1	0	1	1	0	0	1	1

Sum the values where the code is 1:

$$128 + 32 + 16 + 2 + 1 = 179$$

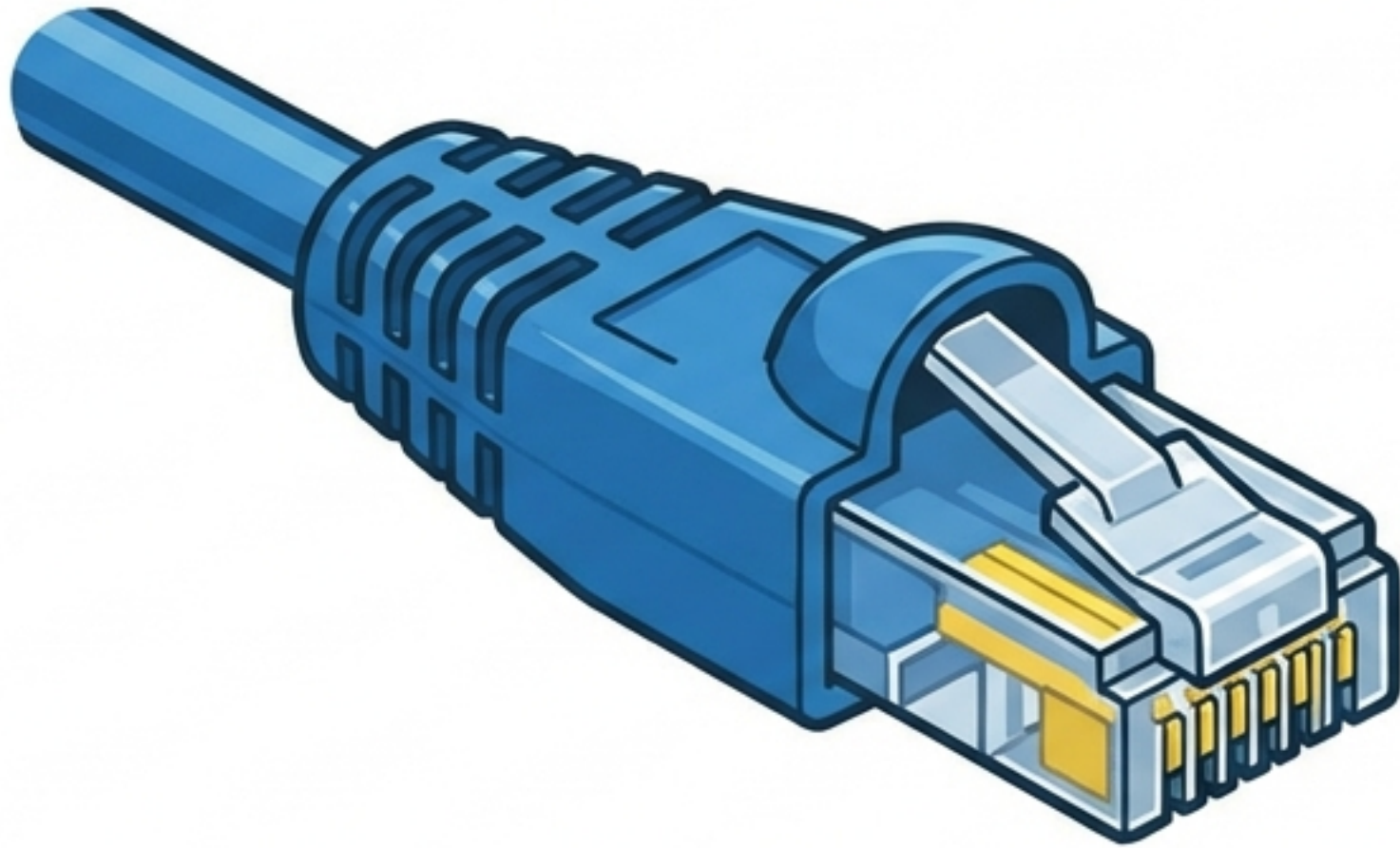
The Highway: Hardware & ISPs



Gateway between
home and network.

Company providing
the connection
(e.g., Verizon, BT).

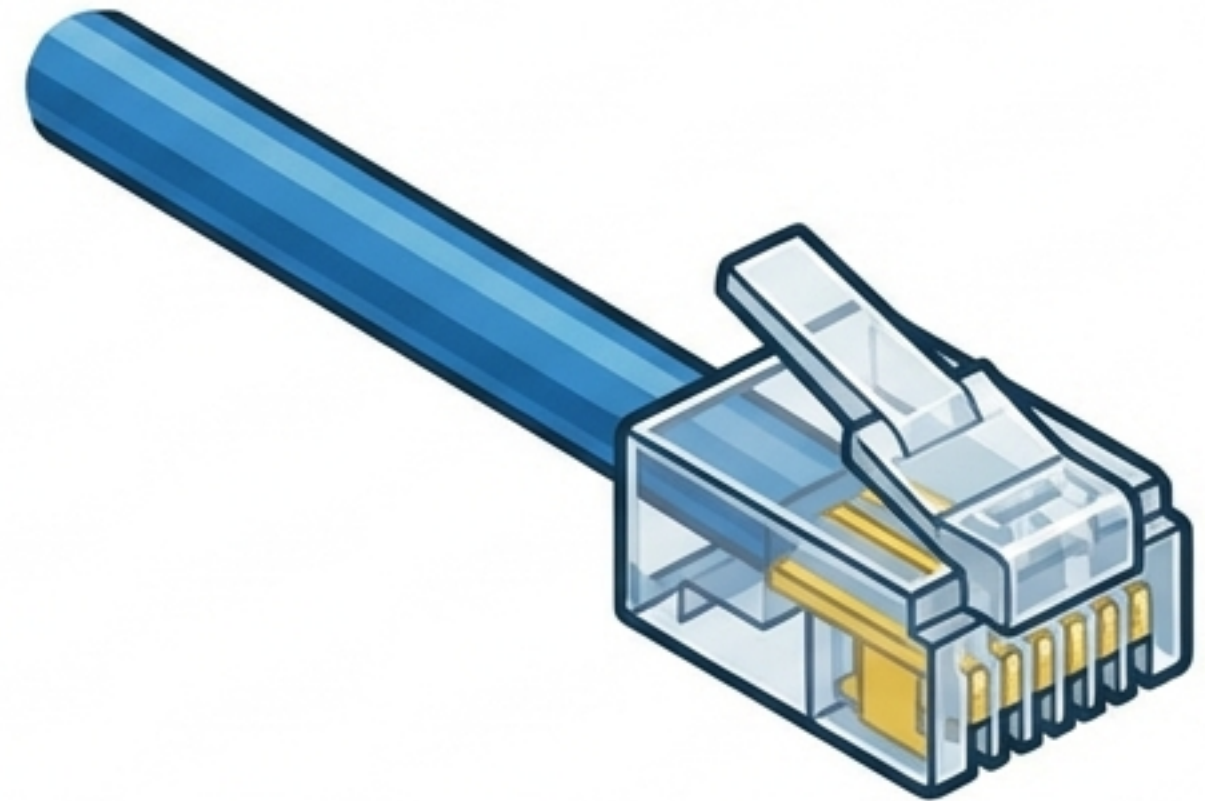
Copper Connections: Ethernet vs. DSL



Ethernet (RJ-45)

Connects PC to Router

Ethernet: High speed, standard for local networks.

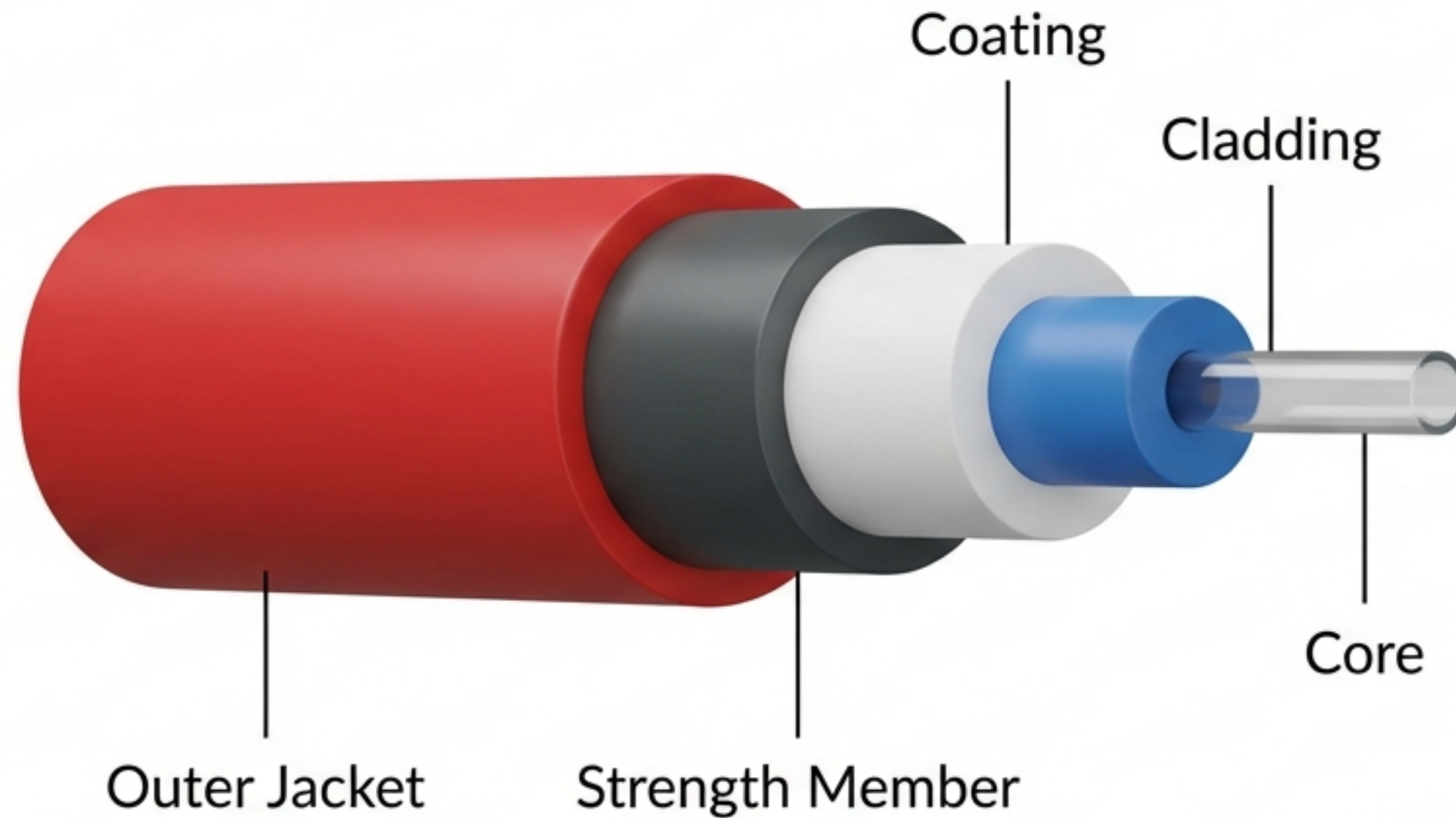


DSL (RJ-11)

Connects Router to Wall (Phone Line)

DSL: Uses telephone lines. Speed drops if you are far from the ISP.

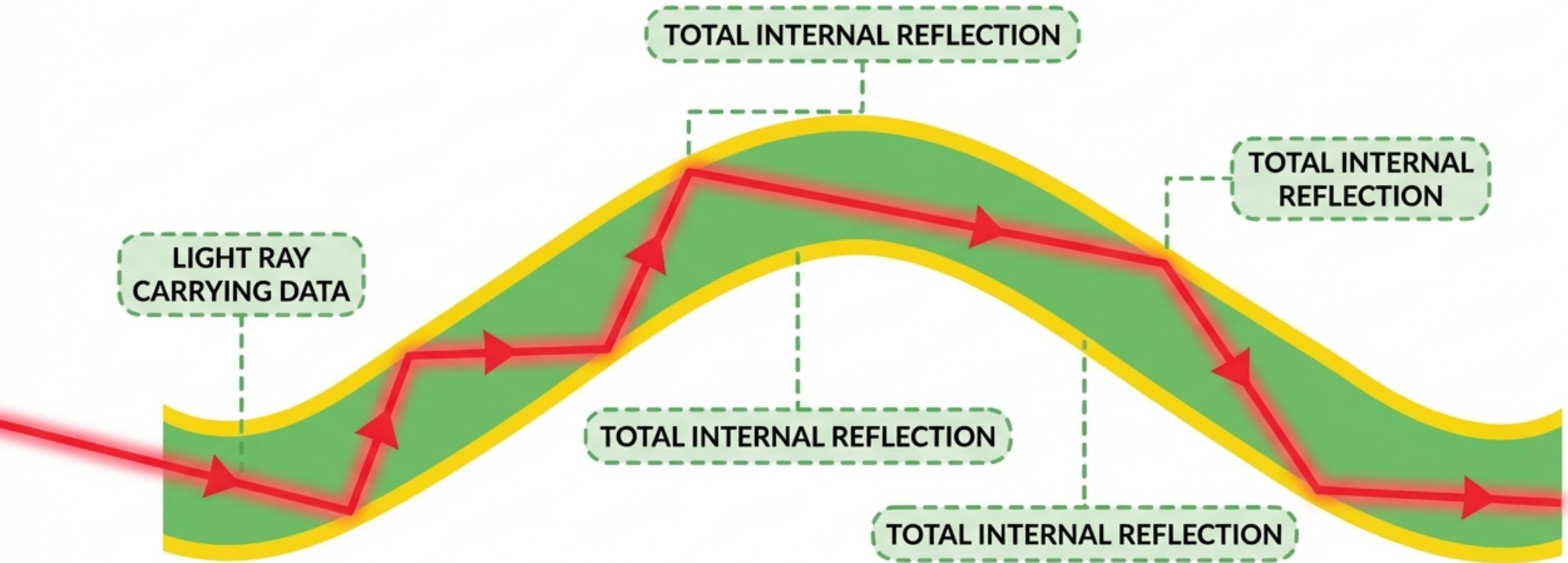
The Speed of Light: Fiber Optics



Transmission: Data travels as pulses of light (LED/Laser).

Speed: The fastest way to send data. The backbone of the modern internet.

Science Spotlight: Total Internal Reflection



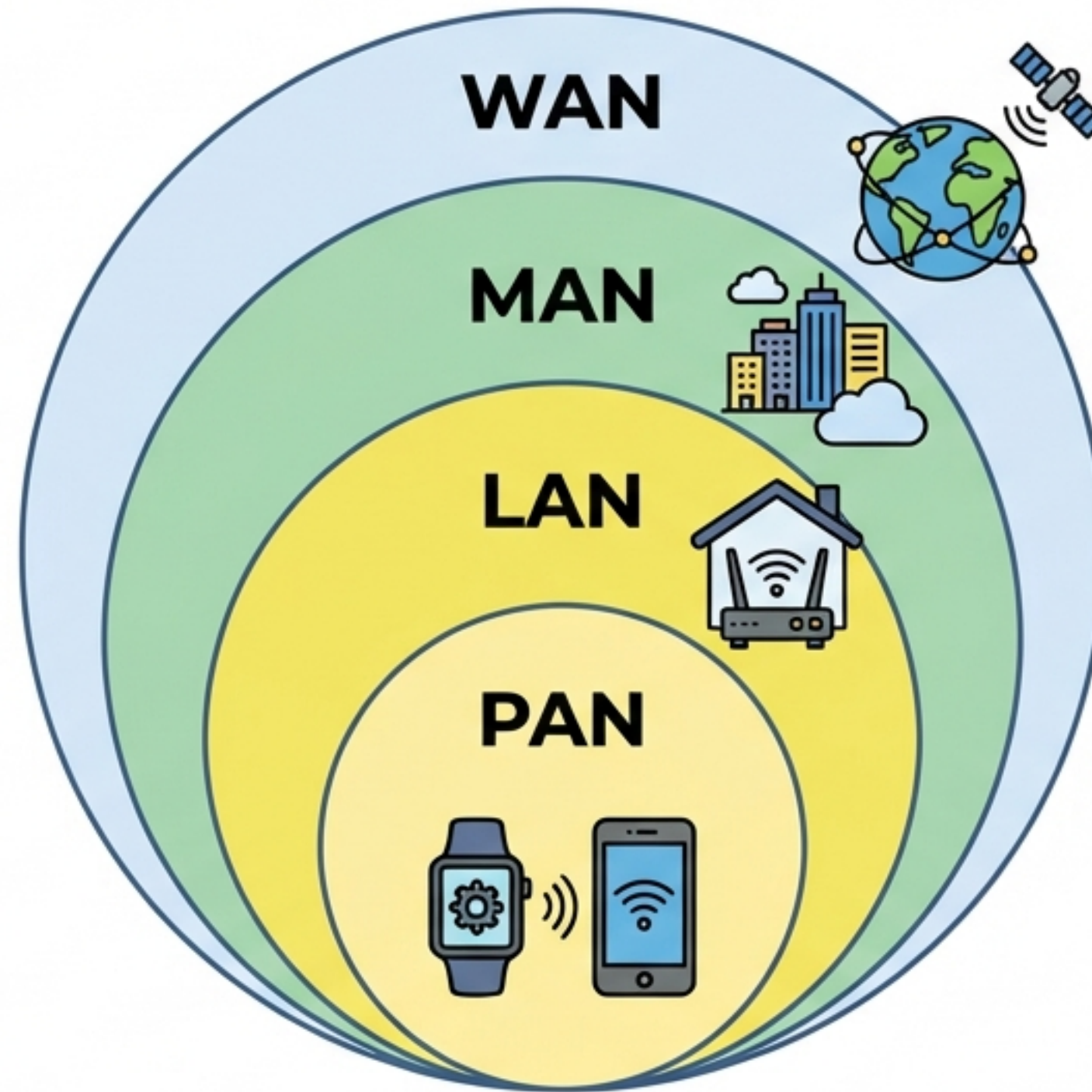
Light reflects and bends inside the glass core without escaping, allowing data to travel around corners.

Cable Showdown: Speed & Cost

Cable Type	Cost	Download Speed
DSL	Low	5-6 Mbps
Ethernet	Medium	2-50 Mbps
Fiber Optic	High	5-10,000 Mbps

Evolution of Hardware: DSL → Ethernet → Fiber Optic

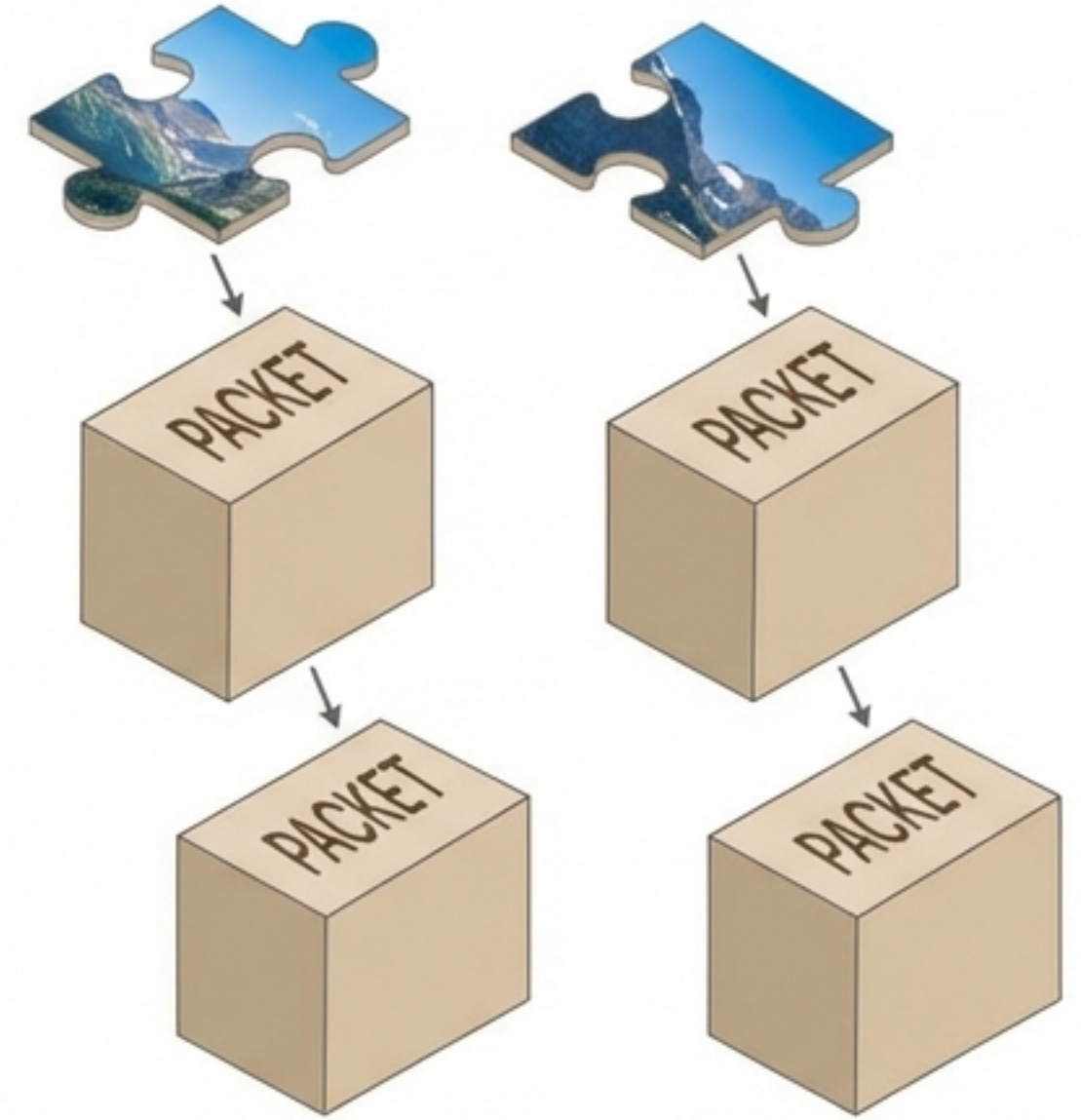
Circles of Connection: Network Scale



Small networks connect to form larger networks, eventually creating the Internet.

Data in a Box: The Packet

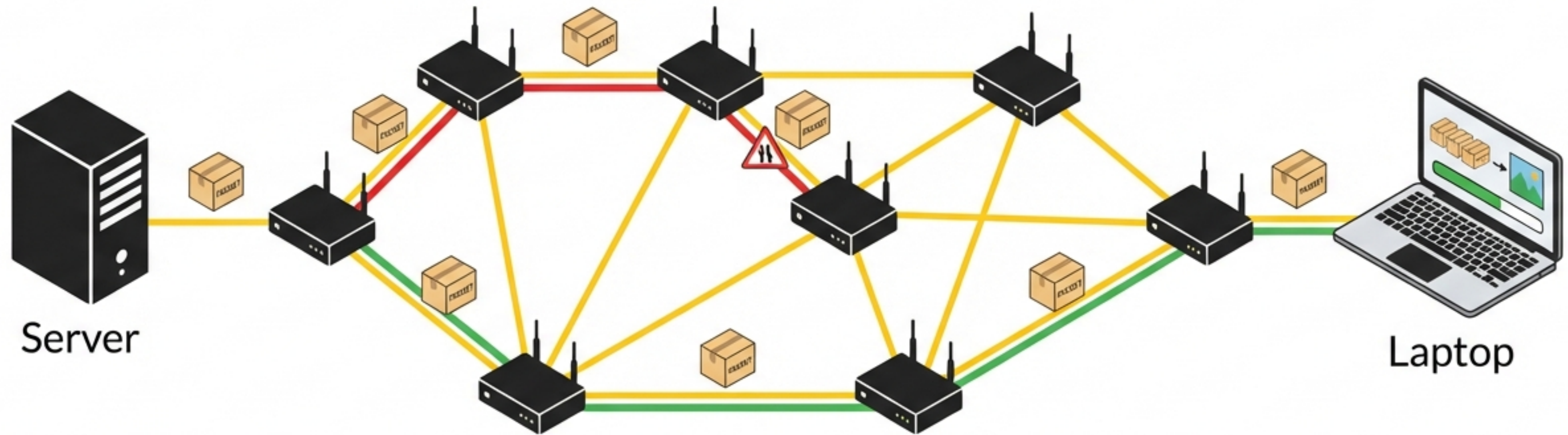
Slicing



The Problem: Files are too big to send at once.

The Solution: Break them into '**Packets**'. Each packet contains the data chunk + the address of where it's going.

The Route & The Reassembly



Smart Routing:

If one path is busy (congested), routers send packets via an alternative route.

Reassembly:

The receiving device collects all packets and puts the puzzle back together.

Tech Flash: Key Vocabulary

ISP: Internet Service Provider (The company connecting you).

Binary: The computer language of 0s and 1s.

Packet: A small batch of data carrying an address.

ASCII: The code that turns letters into numbers.

Fiber Optics: Cables that transmit data using light.



“Millions of packets fly around the world every second, allowing us to access information anytime, anywhere.”